```
MariaDB [(none)]> CREATE DATABASE company;
Query OK, 1 row affected (0.006 sec)

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| DataEx1            |
| company            |
| information_schema |
| mysql              |
| performance_schema |
| sql_kyselyt        |
| sys                |
+--------------------+
7 rows in set (0.001 sec)

MariaDB [(none)]> USE company;
Database changed
MariaDB [company]> CREATE TABLE employee (
    -> emp_id INT PRIMARY KEY,
    -> first_name VARCHAR(40),
    -> last_name VARCHAR(40),
    -> birth_day DATE,
    -> sex VARCHAR(1),
    -> salary int,
    -> super_id int,
    -> branch_id INT);
Query OK, 0 rows affected (0.040 sec)

MariaDB [company]> DESC employee;
```

```
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| emp_id     | int(11)     | NO   | PRI | NULL    |       |
| first_name | varchar(40) | YES  |     | NULL    |       |
| last_name  | varchar(40) | YES  |     | NULL    |       |
| birth_day  | date        | YES  |     | NULL    |       |
| sex        | varchar(1)  | YES  |     | NULL    |       |
| salary     | int(11)     | YES  |     | NULL    |       |
| super_id   | int(11)     | YES  |     | NULL    |       |
| branch_id  | int(11)     | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
8 rows in set (0.016 sec)
MariaDB [company]> CREATE TABLE branch (
    -> branch_id INT PRIMARY KEY,
    -> branch_name VARCHAR(40),
    -> mgr_id INT,
    -> mgr_start_date DATE,
    -> FOREIGN KEY(mgr_id) REFERENCES employee(emp_id) ON DELETE SET NULL);
Query OK, 0 rows affected (0.038 sec)

MariaDB [company]> DESC branch;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| branch_id     | int(11)     | NO   | PRI | NULL    |       |
| branch_name   | varchar(40) | YES  |     | NULL    |       |
| mgr_id        | int(11)     | YES  | MUL | NULL    |       |
| mgr_start_date | date       | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
4 rows in set (0.006 sec)
```

```
MariaDB [company]> ALTER TABLE employee
    -> ADD FOREIGN KEY(branch_id)
    -> REFERENCES branch(branch_id)
    -> ON DELETE SET NULL;
Query OK, 0 rows affected (0.055 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [company]> ALTER TABLE employee
    -> ADD FOREIGN KEY(super_id)
    -> REFERENCES employee(emp_id)
    -> ON DELETE SET NULL;
Query OK, 0 rows affected (0.063 sec)
Records: 0  Duplicates: 0  Warnings: 0
MariaDB [company]> CREATE TABLE client (
    -> client_id INT PRIMARY KEY,
    -> client_name VARCHAR(40),
    -> branch_id INT,
    -> FOREIGN KEY(branch_id) REFERENCES branch(branch_id) ON DELETE SET NULL);
Query OK, 0 rows affected (0.038 sec)

MariaDB [company]> DESC client;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| client_id   | int(11)     | NO   | PRI | NULL    |       |
| client_name | varchar(40) | YES  |     | NULL    |       |
| branch_id   | int(11)     | YES  | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
3 rows in set (0.007 sec)
```

```
MariaDB [company]> CREATE TABLE works_with (
    -> emp_id INT,
    -> client_id INT,
    -> total_sales INT,
    -> PRIMARY KEY(emp_id, client_id),
    -> FOREIGN KEY(emp_id) REFERENCES employee(emp_id) ON DELETE CASCADE,
    -> FOREIGN KEY(client_id) REFERENCES client(client_id) ON DELETE CASCADE);
Query OK, 0 rows affected (0.029 sec)

MariaDB [company]> DESC works_with;
+-------------+---------+------+-----+---------+-------+
| Field       | Type    | Null | Key | Default | Extra |
+-------------+---------+------+-----+---------+-------+
| emp_id      | int(11) | NO   | PRI | NULL    |       |
| client_id   | int(11) | NO   | PRI | NULL    |       |
| total_sales | int(11) | YES  |     | NULL    |       |
+-------------+---------+------+-----+---------+-------+
3 rows in set (0.007 sec)

MariaDB [company]> CREATE TABLE branch_supplier ( branch_id INT, supplier_name VARCHAR(40),
supply_type VARCHAR(40), PRIMARY KEY(branch_id, supplier_name),
FOREIGN KEY(branch_id) REFERENCES branch(branch_id) ON DELETE CASCADE);
Query OK, 0 rows affected (0.027 sec)

MariaDB [company]> DESC branch_supplier;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| branch_id     | int(11)     | NO   | PRI | NULL    |       |
| supplier_name | varchar(40) | NO   | PRI | NULL    |       |
| supply_type   | varchar(40) | YES  |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
3 rows in set (0.007 sec)
```

```
MariaDB [company]> SHOW tables;
+------------------+
| Tables_in_company |
+------------------+
| branch           |
| branch_supplier  |
| client           |
| employee         |
| works_with       |
+------------------+
5 rows in set (0.004 sec)

MariaDB [company]> INSERT INTO employee VALUES(100, 'David', 'Wallace', '1967-11-17', 'M', 250000, NULL, NULL);
Query OK, 1 row affected (0.014 sec)

MariaDB [company]>

MariaDB [company]> INSERT INTO branch VALUES(1, 'Corporate', 100, '2006-02-09');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]>

MariaDB [company]> UPDATE employee
    -> SET branch_id = 1
    -> WHERE emp_id = 100;
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [company]>

MariaDB [company]> INSERT INTO employee VALUES(101, 'Jan', 'Levinson', '1961-05-11', 'F', 110000, 100, 1);
Query OK, 1 row affected (0.003 sec)

MariaDB [company]>
```

```
MariaDB [company]> -- Scranton

MariaDB [company]> INSERT INTO employee VALUES(102, 'Michael', 'Scott', '1964-03-15', 'M',
75000, 100, NULL);

Query OK, 1 row affected (0.002 sec)


MariaDB [company]>

MariaDB [company]> INSERT INTO branch VALUES(2, 'Scranton', 102, '1992-04-06');

Query OK, 1 row affected (0.002 sec)


MariaDB [company]>

MariaDB [company]> UPDATE employee

   -> SET branch_id = 2

   -> WHERE emp_id = 102;

Query OK, 1 row affected (0.004 sec)

Rows matched: 1  Changed: 1  Warnings: 0


MariaDB [company]>

MariaDB [company]> INSERT INTO employee VALUES(103, 'Angela', 'Martin', '1971-06-25', 'F',
63000, 102, 2);

Query OK, 1 row affected (0.004 sec)


MariaDB [company]> INSERT INTO employee VALUES(104, 'Kelly', 'Kapoor', '1980-02-05', 'F', 55000,
102, 2);

Query OK, 1 row affected (0.002 sec)


MariaDB [company]> INSERT INTO employee VALUES(105, 'Stanley', 'Hudson', '1958-02-19', 'M',
69000, 102, 2);

Query OK, 1 row affected (0.003 sec)


MariaDB [company]>

MariaDB [company]> -- Stamford

MariaDB [company]> INSERT INTO employee VALUES(106, 'Josh', 'Porter', '1969-09-05', 'M', 78000,
100, NULL);
```

```
INSERT INTO branch VALUES(3, 'Stamford', 106, '1998-02-13');

UPDATE employee
SET branch_id = 3
WHQuery OK, 1 row affected (0.005 sec)

MariaDB [company]>
MariaDB [company]> INSERT INTO branch VALUES(3, 'Stamford', 106, '1998-02-13');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]>
MariaDB [company]> UPDATE employee
    -> SET branch_id = 3
    -> WHERE emp_id = 106;
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [company]>
MariaDB [company]> INSERT INTO employee VALUES(107, 'Andy', 'Bernard', '1973-07-22', 'M',
65000, 106, 3);
Query OK, 1 row affected (0.003 sec)

MariaDB [company]> INSERT INTO employee VALUES(108, 'Jim', 'Halpert', '1978-10-01', 'M', 71000,
106, 3);
Query OK, 1 row affected (0.003 sec)

MariaDB [company]>
MariaDB [company]>
MariaDB [company]> -- BRANCH SUPPLIER
MariaDB [company]> INSERT INTO branch_supplier VALUES(2, 'Hammer Mill', 'Paper');
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [company]> INSERT INTO branch_supplier VALUES(2, 'Uni-ball', 'Writing Utensils');
Query OK, 1 row affected (0.005 sec)

MariaDB [company]> INSERT INTO branch_supplier VALUES(3, 'Patriot Paper', 'Paper');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO branch_supplier VALUES(2, 'J.T. Forms & Labels', 'Custom Forms');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO branch_supplier VALUES(3, 'Uni-ball', 'Writing Utensils');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO branch_supplier VALUES(3, 'Hammer Mill', 'Paper');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO branch_supplier VALUES(3, 'Stamford Lables', 'Custom Forms');
Query OK, 1 row affected (0.002 sec)

MariaDB [company]>
MariaDB [company]> -- CLIENT
MariaDB [company]> INSERT INTO client VALUES(400, 'Dunmore Highschool', 2);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO client VALUES(401, 'Lackawana Country', 2);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO client VALUES(402, 'FedEx', 3);
Query OK, 1 row affected (0.003 sec)

MariaDB [company]> INSERT INTO client VALUES(403, 'John Daly Law, LLC', 3);
```

```
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO client VALUES(404, 'Scranton Whitepages', 2);
VALUES(405, 'Times Newspaper', 3);
INSERT INTO client VALUES(406, 'FedEx', 2);

-- WORKS_WITH
INSERTQuery OK, 1 row affected (0.019 sec)

MariaDB [company]> INSERT INTO client VALUES(405, 'Times Newspaper', 3);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO client VALUES(406, 'FedEx', 2);
 INTO works_with VALUES(105, 400, 55000);
INSERT IQuery OK, 1 row affected (0.003 sec)

MariaDB [company]>
MariaDB [company]> -- WORKS_WITH
MariaDB [company]> INSERT INTO works_with VALUES(105, 400, 55000);
NTO works_with VALUES(102, 401, 267000);
INSERT INQuery OK, 1 row affected (0.003 sec)

MariaDB [company]> INSERT INTO works_with VALUES(102, 401, 267000);
TO works_with VALUES(108, 402, 22500);
INSERT INTOQuery OK, 1 row affected (0.004 sec)

MariaDB [company]> INSERT INTO works_with VALUES(108, 402, 22500);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO works_with VALUES(107, 403, 5000);
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [company]> INSERT INTO works_with VALUES(108, 403, 12000);
Query OK, 1 row affected (0.003 sec)

MariaDB [company]> INSERT INTO works_with VALUES(105, 404, 33000);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO works_with VALUES(107, 405, 26000);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO works_with VALUES(102, 406, 15000);
Query OK, 1 row affected (0.002 sec)

MariaDB [company]> INSERT INTO works_with VALUES(105, 406, 130000);
Query OK, 1 row affected (0.002 sec)
MariaDB [company]> SELECT * FROM employee;
+--------+------------+-----------+------------+------+--------+----------+-----------+
| emp_id | first_name | last_name | birth_day  | sex  | salary | super_id | branch_id |
+--------+------------+-----------+------------+------+--------+----------+-----------+
|    100 | David      | Wallace   | 1967-11-17 | M    | 250000 |     NULL |         1 |
|    101 | Jan        | Levinson  | 1961-05-11 | F    | 110000 |      100 |         1 |
|    102 | Michael    | Scott     | 1964-03-15 | M    |  75000 |      100 |         2 |
|    103 | Angela     | Martin    | 1971-06-25 | F    |  63000 |      102 |         2 |
|    104 | Kelly      | Kapoor    | 1980-02-05 | F    |  55000 |      102 |         2 |
|    105 | Stanley    | Hudson    | 1958-02-19 | M    |  69000 |      102 |         2 |
|    106 | Josh       | Porter    | 1969-09-05 | M    |  78000 |      100 |         3 |
|    107 | Andy       | Bernard   | 1973-07-22 | M    |  65000 |      106 |         3 |
|    108 | Jim        | Halpert   | 1978-10-01 | M    |  71000 |      106 |         3 |
+--------+------------+-----------+------------+------+--------+----------+-----------+
9 rows in set (0.001 sec)
```

```
MariaDB [company]> SELECT * FROM branch;
+-----------+-------------+--------+----------------+
| branch_id | branch_name | mgr_id | mgr_start_date |
+-----------+-------------+--------+----------------+
|         1 | Corporate   |    100 | 2006-02-09     |
|         2 | Scranton    |    102 | 1992-04-06     |
|         3 | Stamford    |    106 | 1998-02-13     |
+-----------+-------------+--------+----------------+
3 rows in set (0.001 sec)
MariaDB [company]> SELECT * FROM branch_supplier;
+-----------+---------------------+-----------------+
| branch_id | supplier_name       | supply_type     |
+-----------+---------------------+-----------------+
|         2 | Hammer Mill         | Paper           |
|         2 | J.T. Forms & Labels | Custom Forms    |
|         2 | Uni-ball            | Writing Utensils |
|         3 | Hammer Mill         | Paper           |
|         3 | Patriot Paper       | Paper           |
|         3 | Stamford Lables     | Custom Forms    |
|         3 | Uni-ball            | Writing Utensils |
+-----------+---------------------+-----------------+
7 rows in set (0.000 sec)
MariaDB [company]> SELECT * FROM client;
+-----------+---------------------+-----------+
| client_id | client_name         | branch_id |
+-----------+---------------------+-----------+
|       400 | Dunmore Highschool  |         2 |
|       401 | Lackawana Country   |         2 |
|       402 | FedEx               |         3 |
|       403 | John Daly Law, LLC  |         3 |
|       404 | Scranton Whitepages |         2 |
```

```
|      405 | Times Newspaper     |        3 |
|      406 | FedEx               |        2 |
+----------+---------------------+----------+
7 rows in set (0.001 sec)
MariaDB [company]> SELECT * FROM works_with;
+--------+-----------+-------------+
| emp_id | client_id | total_sales |
+--------+-----------+-------------+
|    102 |       401 |      267000 |
|    102 |       406 |       15000 |
|    105 |       400 |       55000 |
|    105 |       404 |       33000 |
|    105 |       406 |      130000 |
|    107 |       403 |        5000 |
|    107 |       405 |       26000 |
|    108 |       402 |       22500 |
|    108 |       403 |       12000 |
+--------+-----------+-------------+
9 rows in set (0.001 sec)
```